Ask Python Docs

A retrieval-augmented assistant for students working with Python. Groq generates concise answers from retrieved official documentation, with numbered source citations.

Corpus	Snapshot
Documentation	Official Python 3.12 tutorial and selected standard-library pages
HTML source pages	73
Sections / chunks	784 / 2632
Evaluation cases	110 answerable + 10 out-of-scope
Intended users	Classmates who use Python; formal user feedback pending

The collection is large enough to require retrieval and contains checkable API behavior. Source URLs and section titles accompany every chunk. Cached raw HTML and a SHA-256 corpus manifest support reproducibility; HTML page count is not a printed-page count.

The evaluation dataset is curated, with symbol-based passage labels resolved automatically. These labels need human review. Development and test questions share topics, so the results do not demonstrate unseen-topic generalization.

Source: https://docs.python.org/3.12/ | License: https://docs.python.org/3.12/license.html

Instructor approval, team-member names and actual contributions, real-user feedback, and a public deployment URL remain external submission tasks.

Pipeline & design

Offline: download -> extract sections -> chunk -> embed -> persist vectors. Online: question -> BM25 + cosine search -> reciprocal rank fusion -> cross-encoder reranking -> Groq generation -> citation validation.

Chunks target 220 whitespace-delimited words with up to 40 words of whole-line overlap. Code indentation is retained where possible. Tables, long lines and nested definitions can lose context, and neural tokenizer limits differ from word counts.

Component	Choice and reason
Vectors	all-MiniLM-L6-v2; normalized vectors in a NumPy matrix. Exact search is sufficient for 2,632 chunks.
Lexical retrieval	BM25 over prose and Python identifiers.
Fusion	Reciprocal rank fusion with constant 60. Rank scores are not confidence probabilities.
Reranker	ms-marco-MiniLM-L-6-v2; baseline 24 candidates, optimized 8.
Generator	Groq openai/gpt-oss-20b, selected from the live model list for this account. Strict JSON claims carry source IDs.
Rejection	Retrieval threshold 0 on reranker logits; generation also abstains when evidence is insufficient.

The app constructs numbered citations from validated IDs. This checks structure, not factual entailment. No tools or commands are executed from the question or source text. Missing keys select an explicit extractive baseline in auto mode; provider errors are shown to users.

Configuration uses a local ignored .env file. Only the question and public documentation evidence are sent to Groq. The model key is never included in browser responses or committed artifacts.

Retrieval evaluation

Benchmark generator: groq. Test retrieval uses 55 answerable questions; answer evaluation also includes 10 negative questions. Recall@5 here means at least one weakly labeled relevant chunk in five results; MRR uses the first relevant rank.

Method	Recall@5	MRR@5	p50 ms	p95 ms
keyword	0.945	0.852	6.1	9.0
vector	0.909	0.772	16.0	19.8
hybrid	0.964	0.842	17.4	25.9
hybrid_rerank	0.945	0.874	92.8	113.8

Development chunk sensitivity; weak labels are remapped for each index. The 220-word default ties the 320-word recall while providing shorter contexts.

Words / overlap	Dev reranked recall@5
140_20	0.855
220_40	0.891
320_60	0.891

All ablations use the same corpus, queries and candidate budget: keyword only, vector only, hybrid, and hybrid plus reranker. Models are warmed before timing. Startup and downloads are excluded. Timings are serial local measurements, not production load-test SLOs.

A relevant passage can mention the API without containing the full answer. Symbol-based source labeling can therefore over-credit retrieval. The saved source passages and retrieved IDs support manual review rather than hiding this limitation.

Chunk-size/overlap sensitivity is reproducible with scripts/chunk_experiment.ps1. A configuration should be selected on development data, then tested on held-out questions. No unrun experiment is represented as completed.

Raw artifacts: data/eval_questions.jsonl; data/corpus_manifest.json; results/groq/retrieval_test.json and metrics_test.json.

Answer quality

Generation is measured separately from retrieval. The automatic correctness proxy requires the expected API symbol and a relevant citation, or appropriate abstention for a negative question. It is not a human correctness score.

Exact-copy support is meaningful only for the extractive baseline. Groq output is evaluated semantically for correctness, claim support and whether each citation supports the associated claim. Citation syntax alone is not called faithfulness.

Profile	n	Correct	Faithful	Citations
optimized	6	1.000	1.000	0.833
baseline	6	0.833	0.833	0.833

Same-model LLM judge for answerable cases; deterministic expected-abstention check for negatives. Not human evaluation. Evenly spaced sample of at most 6 cases per profile; not a full-test score.

The semantic judge uses Groq, potentially the same model that generated the answer. This introduces correlated errors and optimism. Human review should inspect a representative sample and all disagreements, using source evidence as the authority.

A 120-case corpus is supplied, but its test split is 65 cases and its dev split is 55. Negative questions test out-of-corpus behavior. The default threshold was not tuned on the test negatives.

Feedback capture is implemented in the browser. Actual classmates must try their own questions; names, private code and student records must not be submitted. No mock feedback is presented as user evidence.

Optimization & cost

The primary experiment reduces reranker candidates from 24 to 8 while holding the corpus, models, generator, output budget and questions fixed. Response caching is disabled in the paired comparison. Explicit pacing sleeps between calls are excluded; provider backoff inside a request is included.

Profile	Proxy acc.	p50 ms	p95 ms	Tokens
optimized	0.877	1411.2	5451.8	82027
baseline	0.846	1626.6	6930.0	82556

Measured p95 reduction: 21.3%; correctness-proxy loss: -3.08 percentage points. Numeric 50% / 2-point target met: False. Human correctness confirmation is still required.

The first offline run reduced p95 by about 46% without proxy loss, below the 50% goal. Generation and network overhead can dilute local reranking savings. A failed optimization target is reported as a failure rather than adjusted away.

With GROQ_FREE_TIER=true, API cost is zero under the configured free allowance, subject to quotas. The observed account limit was 8,000 tokens/minute. Record actual token usage for extrapolation; do not claim a 50% billing reduction from a zero baseline.

For monthly hosting cost H and N requests, fixed allocation is H/N. At 100x traffic, H/(100N) applies only if that hardware still meets capacity. Add token costs for a paid tier and remeasure queues, throughput and p95. Hardware and electricity are excluded from API-cost fields.

Delivery & lessons

Included: FastAPI API and browser interface; one-command Windows setup; reproducible corpus and neural index; evaluation cases; retrieval ablations; Groq generation; token accounting; citation checks; bounded response cache; tests; Docker build; design, optimization and postmortem reports.

The service serializes local inference to avoid device contention. This limits concurrent throughput. The measured machine has CUDA available; installed models select their device automatically. Normal responses can use an exact-query LRU cache, but repeated-query speedups are not generalized to unseen traffic.

Problems encountered: full disk interrupted setup; a Windows Arrow/torch DLL conflict broke standalone evaluation; the first model was unavailable to the account; loose JSON formatting omitted fields; free-tier rate limits interrupted evaluation. Model discovery, import ordering, strict schemas, bounded retries and progress files addressed these implementation issues.

Known limitations: weak evaluation labels, shared dev/test topics, incomplete table extraction, code truncation, imperfect answerability gates, same-model judgment bias and lack of production load testing. Cross-references can require more passages than the context budget supplies.

Remaining external evidence: instructor corpus approval, independent review, real-user feedback, team contributions, public deployment, and course-specific P1 harness integration. A local URL and Dockerfile do not establish a public deployment.

References: docs.python.org/3.12/; console.groq.com/docs/api-reference; console.groq.com/docs/structured-outputs; sbert.net/examples/sentence_transformer/applications/retrieve_rerank/README.html; fastapi.tiangolo.com/deployment/docker/.

Reproduce: run scripts/setup.ps1; start uvicorn librarian.app:app on port 8000; run pytest; run eval.run for offline and Groq modes; review saved answers; run eval.judge; regenerate the reports. See README.md for exact commands.